## Introduction

Ordinal logistic regression — most commonly the proportional-odds model — handles ordered categorical outcomes such as Likert-scale responses, disease severity grades, NYHA functional class, or cancer stage. It exploits the ordering via a cumulative-probability model: the cumulative odds of being in or below each category share the same set of regression coefficients across all cut-points. Multinomial logistic ignores the ordering and estimates separate coefficients for each non-reference category, which is wasteful when ordering is real and informative.

## Prerequisites

A working understanding of binary logistic regression, ordered categorical variables, and the cumulative-probability framing of ordinal data.

## Theory

The proportional-odds model is

$$\log \frac{\mathrm{P}(Y \leq j)}{\mathrm{P}(Y > j)} = \alpha_j - X^\top \beta, \qquad j = 1, \ldots, k-1,$$

with $k$ ordered categories. The cut-points $\alpha_j$ shift with $j$, but the regression coefficients $\beta$ are constrained to be identical across all cut-points — the proportional-odds assumption. Exponentiated coefficients are cumulative odds ratios: per unit increase in $X$, the odds of being in a higher category multiplied by $\exp(\beta)$, regardless of which threshold is being crossed.

The proportional-odds assumption is testable with Brant's test or with a likelihood-ratio test against a more flexible partial-proportional-odds or generalised ordered logit model.

## Assumptions

Ordered categorical outcome with at least three levels, proportional odds across all cut-points, independent observations, and adequate counts in each category (rule of thumb: at least 5–10 events per coefficient in the smallest category).

## R Implementation

```r
library(MASS); library(brant)

d <- data.frame(
  y = factor(sample(c("low", "med", "high"), 200, replace = TRUE),
             levels = c("low", "med", "high"), ordered = TRUE),
  x = rnorm(200)
)
d$y <- factor(as.character(d$y)[sample(1:200,
                                       prob = 1 + plogis(d$x), replace = TRUE)],
              levels = c("low", "med", "high"), ordered = TRUE)

fit <- polr(y ~ x, data = d, Hess = TRUE)
summary(fit)
exp(coef(fit))

# Test proportional odds
brant(fit)
```

## Output & Results

`polr()` returns coefficients on the cumulative-logit scale plus the cut-point estimates. Exponentiating gives cumulative odds ratios. `brant()` reports an omnibus test for proportional odds plus per-coefficient tests; significant results indicate the assumption fails for that covariate.

## Interpretation

A reporting sentence: "Ordinal logistic regression estimated that each 1-SD increase in $x$ multiplied the cumulative odds of being in a higher category by 2.10 (95 % CI 1.50 to 2.94, $p < 0.001$); Brant's test did not reject the proportional-odds assumption (omnibus $p = 0.44$, all per-covariate $p > 0.20$). The cumulative odds-ratio interpretation applies uniformly across the low-vs-(med + high) and (low + med)-vs-high cut-points." Always test and report the proportional-odds assumption.

## Practical Tips

- Always test the proportional-odds assumption with Brant's test or LRT against a partial-proportional-odds model; reporting cumulative ORs without a check is incomplete.
- When proportional odds fails for one or more covariates, use partial-proportional-odds (`VGAM::vglm` with `cumulative(parallel = FALSE ~ x)`) or generalised ordered logit (`oglmx`).
- For three or four categories with severe non-proportionality, multinomial logistic regression is sometimes simpler than partial-proportional-odds, at the cost of ignoring the ordering.
- `ordinal::clm()` provides a modern implementation with mixed-effects extensions (`clmm`) and richer post-hoc tools.
- Report cumulative odds ratios with confidence intervals; the phrasing "odds of being in a higher category" is the clearest way to communicate the cumulative-OR interpretation.
- For predictions on the absolute-probability scale, use `predict(fit, type = "probs")`; readers absorb predicted category probabilities better than cumulative odds ratios.

## R Packages Used

`MASS::polr()` for the canonical proportional-odds implementation; `ordinal::clm()` for a modern alternative with mixed-effects extensions; `brant` for proportional-odds testing; `VGAM` for partial-proportional-odds and other ordinal extensions; `oglmx` for generalised ordered logit; `gtsummary::tbl_regression()` for publication-ready tables.

## For Reviewers

**What to look for in a paper using this method.**

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models

- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI